



Review article

“Assembly Theory” in life-origin models: A critical review

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ARTICLE INFO

Keywords:

Protolife
Protocells
Abiogenesis
Life origin
Origin of life
Protometabolism
Protocellular metabolomics
ProtoBioCybernetics
Molecular evolution
Chemical evolution
Pre-Darwinian evolution

ABSTRACT

Any homeostatic protometabolism would have required orchestration of disparate biochemical pathways into integrated circuits. Extraordinarily specific molecular assemblies were also required at the right time and place. Assembly Theory conflated with its cousins—Complexity Theory, Chaos theory, Quantum Mechanics, Irreversible Nonequilibrium Thermodynamics and Molecular Evolution theory—collectively have great naturalistic appeal in hopes of their providing the needed exquisite steering and controls. They collectively offer the best hope of circumventing the need for active selection required to formally orchestrate bona fide formal organization (as opposed to the mere self-ordering of chaos theory) (Abel and Trevors, 2006b). This paper focuses specifically on AT's contribution to naturalistic life-origin models.

1. What is Assembly Theory (AT)?

“AT” explores the degree of causation difficulty and minimal requirements needed to produce a given molecular ensemble (Cronin, 2023; Gordon, 2013; Łukaszyk, 2024; Marshall, 2021; Mathis et al., 2024; Miles, 2023; Schwieterman, 2018; Walker, 2024). Lee Cronin and Sara Walker have probably taken the lead in promoting AT (Cardoso et al., 2020; Cronin and Walker, 2016; Liu et al., 2021; Marshall et al., 2021). Sharma et al. (2023) published in Nature is the most formally developed version of AT. Updates, clarifications, additions and defenses have been added mostly within YouTube URLs (Cronin, 2023; Cronin and Fridman, 2024; Mathis et al., 2024; Walker, 2024). Leading critics are Hector Zenil (Zenil, 2023, 2024; Zenil and Tour, 2024), James Tour (2024), Hazen et al. (Hazen et al., 2024), Chemist Steven Benner (2023), and Johannes Jaeger (2024). AT could probably be viewed as a subset or outgrowth of complexity theory (Gordon, 2013; Lambert, 2020), Shannon Uncertainty (entropy) (Shannon, 1948a, 1948b), Kolmogorov Algorithmic Compression Theory (Kolmogorov, 1965, 1983) and LZ compression, Zenil's decade-old block Decomposition Method (Abrahão FS et al., 2024; Ozelim et al., 2024; Uthamacumaran et al., 2024; Zenil, 2023, 2024). The Chaos Theory of self-ordering dissipative structures researched initially by Prigogine and Nicolis also plays a major role (Glansdorf and Prigogine, 1971; Nicolis and Prigogine, 1977, 1989; Prigogine, 1961, 1962, 1967, 1980, 1997; Prigogine et al., 1969, 1972a, b, 1977; Prigogine and Nicolis, 1971; Prigogine and Stengers, 1984).

Behind AT often lies the pre-assumption of impressive creativity producible by irreversible nonequilibrium thermodynamics (Prigogine, 1962; Prigogine et al., 1972a; Torrenegra-Rico et al., 2022). Onsager's Nobel Prize winning work on “reciprocal relations” revealed these spontaneous paired reverse flux disequilibria (Onsager, 1931, 1968). Proponents of irreversible nonequilibrium thermodynamics contend that quanta dispersal over many degrees of microscopic freedom alone is sufficient to explain life origin) (Branscomb, 2019; Hernandez and Michaelian, 2022; Michaelian, 2017a, 2017b, 2018, 2019, 2021, 2022; Morales and Michaelian, 2020). Thermal energy/mass conservation is believed to be maintained by these reciprocal paired fluxes. One flux is thermodynamically downhill, but drives another flux in close proximity uphill. Thermodynamic probabilism is viewed as an “imperative,” as though it were a physicyodynamic cause of creative physical effects. Irreversible far-from-equilibrium fluxes and states are imagined to spontaneously occur. This is viewed as the avenue to spontaneous abiogenic potential. It is hoped that this avenue would obviate the need for any active selection (Abel, 2009c, 2011f, 2024a, c) required for molecular evolution.

Proponents of AT begin with what seems to be indistinguishable from Shannon's uncertainty measures of future possibility space. What is the phase space of physicyodynamic possibilities (Sharma et al., 2023)? Also prominent in AT models is a presumed history of such possibilities having actually occurred.

Many AT proponents hope that AT will help elucidate plugging

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<https://doi.org/10.1016/j.biosystems.2024.105378>

Received 17 August 2024; Received in revised form 30 November 2024; Accepted 4 December 2024

Available online 20 December 2024

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abiogenesis dilemmas in an inanimate environmental context. Abiogenesis science has the wonderful benefit of providing very helpful reductionism to a lot of very sticky problems. The result is a great deal of simplicity and clarity in studying how steering, control, conceptual complexity, and homeostatic metabolism might have first been accomplished in an inanimate environment.

Of particular interest would be what the molecular assemblies might be able to DO if spontaneously formed. The field of protocellular metabolomics requires an extraordinary degree of molecular functionality not only of individual moieties, but of their collective efficacious pathways to biofunction, their holistic cooperation and integrated circuits. AT focuses on *measuring* the *difficulty* of molecular assemblies. Understanding the contributions of molecular/chemical/pre-Darwinian selection and evolution also plays a crucial role.

If there were any key to understanding “self-organization” and “emergence,” it would come through a synthesis of AT, Complexity Theory, Chaos Theory, Quantum Mechanics and Irreversible Non-equilibrium Thermodynamics. Molecular Evolution is added to the mix in hopes of providing the needed missing explanation of selection for “usefulness.” But the focus of this paper is primarily on AT specifically as it relates to abiogenesis theory.

2. Limitations and shortcomings of relevance to abiogenesis

2.1. Physiochemical and other physicydynamic limitations of AT

Of great interest to abiogenists is AT's assertion that it can distinguish between, and even measure, the difficulty level of spontaneous formation of biotic vs. abiotic molecules. There does appear to be some measurable minimal differences between the two, although the bulk of this difference appears to be related mostly to mere molecular length and size more than to potential biotic function (Abrahão FS et al., 2024; Ozelim et al., 2024; Zenil, 2023; Zenil and Tour, 2024).

No measurements of the difficulty of spontaneous assembly of biochemical pathways, integrated circuits, and homeostatic metabolism are offered by AT.

The question of measurement will be addressed more fully below in section 2.5. For now, the question is, “How did inanimate physicydynamics specifically pursue and produce biotic players, formally organize disparate biochemical pathways, and orchestrate all of these independent biochemical pathways into holistic homeostatic metabolism?” In all fairness to AT proponents, their focus is primarily on the formation of individual biotic molecular assemblies. But how would AT researchers know which molecular assemblies were “biotic” except in the larger context of at least an envisioned theoretical protometabolism? Not all organic molecules are biotic. To do this, they “reconceptualize” in their minds ordinary physical chemistry in order to imagine abiogenic scenarios. The limitations of this human mentation relevant to what actually happened historically and ontologically in a prebiotic environment will be addressed below in Section 2.6.

The question still remains, “What *steered* inorganic and organic chemistry toward abiogenesis in an inanimate environment?” Is the contention that no active selection, steering or controls were needed to produce conceptually complex biotic molecules and their orchestration into biosystems? Or is the contention that mere constraints and laws were adequate to do this active selection, steering and formal control? Steering toward function and controlling that function to maintain or optimize it are formal enterprises, not physicydynamic natural-science phenomena. Neither constraints nor laws are capable of *steering* physicydynamic interactions toward “usefulness.”

A few simple inorganic catalysts could have contributed spontaneously early on. But most required catalysis would have necessarily been protein enzymes having ingenious secondary and tertiary structures. Crude catalytic agents would have to be legion and all-too-specific for many essential protometabolic molecular assemblies to form. Even if they all formed, they would have to be carefully sequenced in their

synthetic reactions at the right times and under the required conditions. Reaction sequences would have to be selectively interrupted just at the right time to preclude biochemical catastrophe. Any protometabolism would have to be formally orchestrated, not merely self-ordered (Abel, 2024b; Abel and Trevors, 2006b), to achieve the slightest hope of homeostasis.

What determined the sequential polymerizations (primary structure) that *only later* caused the effect of the needed secondary and tertiary structures? Where did all the needed specific chaperones come from to assist folding? How did these chaperones just happen to be waiting at the ribosomal exit port for each emerging polypeptide? How did the incredible ribosome super-machine come into existence in the first place? How did this machine know how to translate specific polynucleotide sequencing into polyamino acid sequencing? How did all of the t-RNAs and aminoacyl tRNA synthetases/transferases, etc. get structured and then integrated into an orchestrated symphony? All of this had to be waiting at the same time and place in order to even define what would be “biogenic” individual molecular assemblies.

Many more problems exist.

- What about the fact that many potential players in abiogenesis models have very short half-lives?
- What about the homochirality problem? Yes, Sharma (2023) has recently published a very interesting paper on homochirality origin. But this model is quite limited in scope.
- What about the spin-polarized electron problem (Ozturk and Dimitar, 2022)?
- Does mere molecular stability equal molecular function?
- Wouldn't a self-replicating RNA-peptide assembly tend to consume all the resources needed for the formation of many other assemblies required for protolife to “self-organize”?

Biochemical reality demands acknowledgement of the inadequacy of physicydynamics alone to orchestrate life. The only thing preventing the production of chemical “tar” is precise prescription of many synthetic steps and factors specified in great detail by Prof. James Tour at Rice University (Thaxton et al., 2020; Tour, 2020, 2024). Tour's Harvard debate with Lee Cronin expounds on these severe biochemical limitations (Zenil and Tour, 2024). Lee Cronin is probably the number-one proponent of AT (Cronin, 2023; Cronin and Fridman, 2024; Marshall et al., 2021; Sharma et al., 2023).

AT supporters place great emphasis on “possibilities” and “eons of time.” But time is not a cause of any effect. Absolutely no “history” of successful “possibilities” is provided by AT authors wherein spontaneous generation of even protolife is produced. Nothing is more formally organized than homeostatic metabolism far from equilibrium. The only thing “emerging” in the history of human observation seems to be the vivid imagination of the AT theorists' wish fulfillment.

Chemist Steven Benner (2023) contends that AT has not solved “the wickedly complex problem central to organic chemistry: The problem of reactivity.”

Kondratyeva et al. (Kondratyeva et al., 2022) point out the *mutual* need of both proteins and nucleic acids. Co-evolution models suggest a possible physico-chemical and structural link between nucleic acids, proteins and amino acids. They have therefore received great attention (Copley et al., 2007; Guimaraes, 2004, 2010, 2011, 2014, 2015, a, b, 2017, 2019; Guimaraes et al., 2008; Lahav, 1991, 1993; Stevenson, 2002; Tagami and Li, 2023; Wong, 1975). But these models are highly conjectural with limited empirical support. Imagination tends to be very stretched and the envisioned products quite extensive. No steering or controls are offered to justify presupposing the processing pathways needed. Without active selection, this author finds co-evolution models to be theoretically simplistic and naïve.

In the RNA world hypothesis, the emergence of translation was preceded by a hypothesized era of self-replicating RNAs. The problem is, no RNA chemist has ever been able to synthesize ribozymes with

processive polymerase activity (Kondratyeva et al., 2022). If highly skilled RNA chemists cannot manufacture such RNA's, questions are raised about their spontaneous formation according to AT. One of AT's main theorists is an abiogenist whose primary interest is the level of "difficulty" involved in synthesizing biogenic molecular "assemblies" (Sharma et al., 2023).

Useful co-enzymes, peptides, proteins, small RNA's and polynucleotide genes could all be considered highly functional molecular "assemblies." How was the formation and measurement of difficulty of these high-achieving assemblies accomplished? The naturalistic explanation is necessarily limited to mere chance and necessity.

2.1.1. Are chance and necessity sufficient?

A simple piece of wire can be used for many functions. But how many pieces of any kind of wire have ever spontaneously "self-organized," or "emerged" in the history of human observation? Even a simple piece of physical wire is an instantiated formalism. The characteristics of this "molecular ensemble" and physical "object," as AT authors would call it, have never been observed to spontaneously generate from the four known forces of physics, from law, constraints, from chance, or from irreversible nonequilibrium thermodynamics. Iron ore in the ground lacks the characteristics of a long piece of metal alloy with constant diameter and optimized tensile strength and malleability needed to make a simple paper clip.

A piece of wire is an object that shows evidence of active selection in its formation. We do not attribute that selection to iron ore itself, or to the laws of physics and chemistry. An orchestrating, conceptualizing agent did the active selecting that brought this alloy "object" into existence. This "molecular ensemble" was "assembled" as an engineering project, not as a mere physico-chemical sequence of spontaneous "possible" interactions. No empirical history exists of a simple piece of wire "emerging" by "self-organization." Why not? If AT cannot explain the origin of a simple piece of wire, it is not going to explain the origin of life.

Spontaneous formation of innumerable needed life-origin players is statistically prohibitive (Axe, 2000, 2004). Axe found that only one polypeptide in 10^{77} stochastic ensembles would spontaneously fold into a needed tertiary structure. Not only that, but the spontaneous formation of many needed molecular assemblies has Universal Plausibility Metrics of $\xi < 1.0$. The Universal Plausibility Principle (Abel, 2009d, e, 2011i) requires peer-review rejection of any model with that low a quantification of the Universal Plausibility Metric of ξ . The proposed spontaneous formation and collection of so many needed statistically-prohibitive "assemblies" in the same place at the same time is simply scientifically untenable as a "natural" phenomenon.

2.1.2. Constraints vs. controls

Constraints are not Controls (Abel, 2010, 2024b, c). Mere constraints cannot steer and control events toward conceptual, orchestrational, integrative, formal function.

"The unreasonable effectiveness of mathematics in the natural sciences" (Hamming, 1980; Wigner, 1960) is only unreasonable for those who fanatically refuse to acknowledge the third fundamental category of reality: Choice Causation (Abel, 2007, 2008a, b, 2009a, 2010, 2011d, e, f, h, j, k, l, 2012, 2015, 2024c; Abel and Trevors, 2005, 2006b; D'Onofrio et al., 2012a; Trevors and Abel, 2004). No rational basis or empirical experience exists to justify attributing homeostatic metabolism to Chance and Necessity alone.

The self-ordering phenomena of chaos theory are usually grossly overextended into wild imaginations of "self-organization" and "emergence" of formalisms. The self-orderings of chaos theory are not only blind to "usefulness," they usually destroy usefulness. Consider weather fronts, hurricanes and tornadoes. How much organization is produced by these self-ordered "dissipative structures"?

Controls require active selection prior to the realization of the function produced by those active selections. The selections cannot be

passive and secondary to eventual functionality. Any needed function that might be favored is only later produced. Even then, inanimate nature does not favor function over non-function. The environment only favors the fittest already-living organisms.

Sophisticated functionality must be pursued with purposeful choices. Such conceptually complex utility has never been observed to be produced by mere constraints or law. The notions of "self-organization" and "emergence" violate this empirical reality. These two terms have no place in natural science. "Self-organization" is a nonsense term. Nothing can possibly organize itself into existence. It would have to already exist to organize anything. If it already existed, it could not possibly organize itself into existence. No excuse remains for continuing to appeal to "self-organization" or "emergence" as a physical cause within quality naturalistic science.

All known life is programmed and cybernetically processed (Esposito et al., 2024). Life is formal computation. Constraints cannot cause formal computation. Only the controls requiring active selections can generate formal sophisticated function. No function is as sophisticated as biofunction. Objects and their spontaneous physicochemical combinations do not program or compute. Mere spontaneous molecular assemblies ("combinations") do not produce programs or Turing machines, let alone ribosomes.

Evolution would not have existed yet in an inanimate environment. Nothing was alive yet to differentially survive and reproduce. Any selection in molecular evolution would have been active, not passive and secondary (Abel, 2009c, 2024a). Thus, abiogenesis had to have been steered by controls, not by fixed laws or constraints. If the latter did the programming, the program would consist of all "1's," or of all "0's," by law! "Does not compute!"

In criticizing AT, Jaeger (2024) reveals his own confusion in his clause, "'higher level emergent causal constraints' in computational worlds made of basic objects and their combinations." What exactly are these magical "higher level constraints" that are just pre-supposed metaphysically to have so much formal control capability? In what sense can any physicyodynamic constraint cause "higher level" non-trivial sophisticated formal function? Is this science, or science fiction? Where is the empirical evidence of mere constraints being capable of formal controls? The only time constraints can serve as controls is when those constraints (usually starting conditions) are chosen by investigator involvement in experimental design.

2.1.3. The problem of active selection

AT authors state that, "The assembly is defined as 'the total amount of selection necessary to produce an ensemble of observed objects.'"

Since the object or ensemble of objects forming does not exist yet in completed form, the selection required to achieve it is active, not passive. Evolution is only passive selection. It favors by differential survival and reproduction the fittest already-living organisms after-the-fact of their being alive and fittest.

Active selection is performed "in order to. ..." (realize some goal). That is called Engineering, not "natural" science! One can find the phrase "in order to..." as Freudian slips in hundreds of abiogenesis papers. The authors exhibit total blindness to the fact that this magical phrase completely violates naturalistic presuppositions in favor of advocating intent and goal. They are only unknowingly affirming the third fundamental category of reality: Choice Causation (Abel, 2007, 2008b, 2009b, 2010, 2011b, h, 2012, 2024a; Abel and Trevors, 2005, 2006b; Trevors and Abel, 2004).

Physico-chemical interactions and reactions are simply not capable of active selection in pursuit of functional ensembles. Function is no better than non-function to an inanimate environment. Laws and constraints could care less whether anything functions.

Selection comes into play much earlier than initially envisioned by proponents of the all-sufficiency of irreversible nonequilibrium thermodynamics, Complexity Theory, Chaos Theory and AT. Even if spontaneous physicyodynamic formation of biotic molecules is presupposed,

one ensemble needs to be preferred over others to isolate and promote the continued existence and production of worthwhile ensembles. What would be the impetus of this in a prebiotic environment where no molecule is perceived to have value?

2.1.4. Synthetic organic chemistry requires very careful controls

The components of synthetic chemistry success must react in the right order at the right temperature, pressure, light, and in the right precise quantities with the needed pauses in reactivity to produce anything other than useless tar (Thaxton et al., 2020; Tour, 2020, 2024). Not only does formation assembly difficulty become an issue, but also half-lives.

The carefully ordered steps and conditions required for synthetic chemistry and orchestration of any protolife have been extensively covered in a 2024 paper just published entitled, “Why Is Abiogenesis Such a Tough Nut to Crack” (Abel, 2024c) We will not rehash here all of that relevant material.

2.2. Functional molecule and integrated circuit limitations of AT

There would be little point to pursuing AT if each ensemble of interest did not possess some non-trivial utility. The definition of “useful” should also be isolated from the valuation of human agents. The question is, could the molecules produced as described by AT serve any stand-alone ontological purpose within an inanimate environment? How would the environment have recognized, cared about, or marshalled any usefulness? How would these spontaneous molecular assemblies have been preserved and accumulated? How were these individual molecular assemblies orchestrated into a protometabolism?

Can physicyodynamics, including irreversible nonequilibrium thermodynamics, actually orchestrate and organize formal functions? The ultimate in functionality is found in biofunction, even at the subcellular level of the simplest known organisms. But the problem for AT begins at a much simpler level than that. How well does AT address the source of any kind of non-trivial spontaneous pragmatism?

Physicyodynamics is blind to function. Inanimate nature could care less whether anything “works” in a formal sense. The “work” of physics has nothing to do with utility.

AT can measure the difficulty of spontaneous molecular formation. But can it measure the difficulty of fashioning highly functional molecules, biotic and biogenic molecules especially. Doing so requires presupposing life rather than explaining its formation using nothing but isolated molecular assembly measures. The life that is just presupposed axiomatically is highly programmed, cybernetically processed, and computational. All of these features of life are formal, not physicyodynamic. Formalisms alone generate the potential bio-functionality of any molecular assembly. The bottom-up approach to abiogenesis lacks all empirical accountability. “Self-organization” and “emergence” of formal functions are not observed. They are both just blindly believed. We can’t even define “biotic” outside of the presupposed context of formal life. Organic molecules are not necessarily biotic or biogenic.

Zenil et al., in particular, challenges whether AT can differentiate between biotic and abiotic molecular assemblies, and the difficulty of their formation. (Abrahão FS et al., 2024; Ozelim et al., 2024; Zenil, 2023; Zenil and Tour, 2024).

What exactly was the *natural* cause of the effect of the orchestration of even a theoretical homeostatic protometabolism? To just keep repeating “self-organization” and “emergence” over and over is nothing more than pulling a rabbit out of a hat.

Why would a prebiotic environment care whether anything functioned? What would have been the source of any motivation to prefer or pursue function?

- What were the means of such an achievement?
- Has chance and necessity ever produced sophisticated functions?
- Why would any accidental function have been preserved?

- Exactly how does AT differentiate between functional vs. nonfunctional molecular assemblies?
- Is there any natural process directionality toward the formation of useful assemblies? Or is the usefulness of any assembly purely happenstantial?
- Will AT elucidate the purely physicyodynamic causative route to useful assemblies as opposed to non-useful assemblies?
- How and why would any spontaneous usefulness be preserved and expanded upon by an inanimate environment?
- How many molecular assemblies with happenstantial sophisticated function would AT authors expect to spontaneously form in the same place at the same time?

2.3. Evolutionary limitations of AT

Given enough eons of time, AT proponents believe that laws, constraints, large phase spaces of possibilities, and thermodynamics can produce formalistic physical entities and systems. Formalisms include sophisticated functions normally associated only with engineering. Evolution is claimed to polish the result of these non-evolutionary, purely thermodynamic and physicyodynamic products.

The first problem with this perspective, as discussed above, is that physicyodynamics itself provides no directionality toward the formation of useful assemblies or their holistic cooperation (Abel, 2006, 2007; Abel and Trevors, 2005; Trevors and Abel, 2004). Sooner or later, evolution has to be included in AT’s causative mix. AT does attempt to include evolution in its model.

Neo-Darwinian Evolution is nothing more than the differential survival and reproduction of the fittest already-programmed, already-cybernetically processed, already-living organisms. Neo-Darwinian evolution is after the fact—altogether passive and secondary to phenotypic benefit. Natural selection is incapable of active selection at the programming level in pursuit of eventual utility via computational halting (The Genetic Selection (GS Principle (Abel, 2009c, 2011f)). Evolution has no goal. Neo-Darwinian evolution, therefore, is irrelevant and worthless to abiogenesis research. If evolution is going to be of any help at all, it will have to be molecular/chemical pre-Darwinian evolution.

Even molecular evolution requires selection to optimize function. It has been shown that selection in molecular evolution must be active selection, not secondary passive selection as seen with “natural selection” (Abel, 2006, 2007, 2024a; Abel and Trevors, 2005; Trevors and Abel, 2004). Primary, secondary and tertiary molecular structure must be actively selected prior to the realization of any utility, not secondarily favored because of their only later-realized pragmatic contribution. For example, nucleosides must be individually actively selected for polymerization into a certain sequence of rigidly-bound 3’5’ phosphodiester bonds in order for that sequence to prescribe the transcription and translation of needed proteins. That is active selection, not secondary passive selection.

The problem is, no known basis for active selection in pursuit of function exists in an inanimate naturalistic environment. Little else exists that could even hopefully explain the spontaneous orchestration of any formal organization. Empirical evidence is painfully lacking of quanta dispersal over many degrees of microscopic freedom producing a single sophisticated protein (Axe, 2000, 2004), or orchestrating the slightest hint of formal function (Abel, 2007, 2008a, b, 2009b, c, 2010, 2011a, c, d, e, f, h, k, l, 2012, 2015; Abel and Trevors, 2005, 2006b, 2007; Durston et al., 2007; Durston and Chiu, 2011; Trevors and Abel, 2004).

Even “trial and error” quests are deliberate searches. Trial-and-error quests are motivated by intent, purpose and goal. Neither molecular nor neoDarwinian evolution has any goal.

Just being far from equilibrium does not necessarily produce functionality, let alone biofunction or homeostatic metabolism. And how could a prebiotic environment even identify functionality?

How were their primary structures sequenced so as to produce instructive value in a completely different language of polyamino acid primary structure sequencing? How were secondary and tertiary structures anticipated when primary structure formed? Efficacious prescriptive sequencing was required during pre-function polymerization. That required active selection, not passive after-the-fact selection. And it's not just protein genesis. Thousands of CircRNA's had to have been programmed for regulation.

Molecular evolution theory has no basis for selection other than greater molecular stability and self-catalysis of massive quantities of one entity (Abel, 2009c, 2011f, 2024a, c). Molecular stability has nothing to do with utility. Some of the most toxic molecules to life are the most stable molecules (e.g. heavy metals). Massive self-replication of one RNA analog would deplete the environment of the resources for hundreds of other needed molecular entities to form.

2.4. Cybernetic limitations of AT

As established in section 2.1.2 and 2.1.3, a more appropriate use of the term "selection" is "active selection." Choices must be made in the pursuit of utility that make the realization of that utility possible only after that active selection is made. The programming of computation is a classic example. So is the active selection of the next nucleoside to polymerize in a prescriptive programming string of nucleic acid. Biology as science cannot evade the question of how certain nucleosides are polymerized in a certain order so as to prescribe protein primary, secondary and tertiary structure. All of the chemical bonds are the same 3'5' phosphodiester bonds. Any physicochemical constraint or law would produce a homopolymer rather than genetic *code*. The very key to genetic coding is the lack of physicydynamic propensity in nucleoside selection.

In the absence of active selection, empirical evidence of any sophisticated function spontaneously arising from irreversible nonequilibrium thermodynamics is virtually nonexistent. Claims to the contrary can invariably be shown to have had "investigator involvement" in their experimental design.

Three fundamental categories of reality exist, not just Chance and Necessity (Abel, 2011h). Causation of active selection cannot be physical (Abel, 2011k). It arises from the far side of the Cybernetic Cut (Abel, 2008b; Abel, 2011b). The Cybernetic Cut is the great divide between Physicydynamic Causation ("necessity") and formal Choice Causation. The decision-node programming choices of coded genetic prescription can arise only from the far side of the Cybernetic Cut. Physicydynamics has no ability to program computation or to orchestrate formal organization.

AT authors contend that, "For the shortest path to success, the assembly space captures the *minimal memory*, in terms of the minimal number of operations necessary to construct an object based on objects that could have existed in its past." (Sharma et al., 2023).

What determined the minimal number of operations necessary? Some law of physics? One of the four known forces? Do thermodynamic "possibilities" determine maximum cybernetic efficiency and sophisticated success?

Only active selections at bona fide decision nodes produce the needed synthetic chemistry procedural steps. The authors are searching within their "assembly pool" for the minimal number of steps to computational success. Would this not be the essence of a cybernetic rather than a natural science problem? How did the laws of physics recognize or appreciate such an ideal goal? How did an inanimate environment ascertain what the optimized "assembly pool" steps should be? How did nature overcome Church's and Turing's halting problem (knowing how in advance to successfully compute an optimized solution) (Church, 1936; Turing, 1936)?

Say AT authors, "Once a pathway to assemble an object is discovered, the object can be reproduced." (Sharma et al., 2023) What exactly was the "discoverer" in a prebiotic environment? How did an inanimate

environment know what to do with any gain even if it could recognize one? What was the "knower" of this gain? How was this pathway (justifiably called a "program" or "optimized algorithm") "discovered"? Worse yet, discovery is not causative. It is an after-the-fact purely epistemological gain. Abiogenic science is interested in *causes* of real effects.

Did physicydynamics have some search criteria? Did the inanimate "objects" go looking to discover an optimized pathway to assembly? Or did all this just come together at the same place and time within eons of time? What is this ruse we keep proudly proclaiming to be "science"?

The quest of explaining "orchestration of formal systems" turns clearly cybernetic, as freely acknowledged by AT authors themselves, when they introduce the question of "What would be the fewest steps needed to accomplish the goal?" (Sharma et al., 2023) AT authors also ask the purely formal, cybernetic question, "What possible pathway would require the *shortest processing time*?" (Sharma et al., 2023) These are cybernetic questions, not natural science questions.

The problem is that cybernetics by definition pertains to *control*. Controls are not constraints (Abel, 2010). Controls require purposeful choices. Controls steer interactions toward formal success. Systems Biology is impossible without controls. We are shooting ourselves in the foot by disallowing discussion of steering and the needed formal controls when trying to elucidate the origins of Systems Biology. We are looking in the wrong place. Constraints are blind and indifferent to "usefulness." We are looking to fixed *laws* when we should be looking to formal programming *rules*.

Bona fide organization lies in the engineering domain, not the physical science domain. The great hope of AT is that it could offer a solution to this naturalistic/physicalistic dilemma. The goal of naturalistic science has always been to explain the origin of exquisite biological orchestration and organization without dependence upon formal choice, steering or controls. Our great white hope has always been "eons of time." Time is not a cause of any effect! Our modus operandi has always been to replace any hint of "teleology" with "teleonomy" (Bartlett, 2023a; Mayr, 1965; Pittendrigh, 1958; Reese, 1994). What's the difference? The answer is that teleonomy tries to disallow the required active selection. We are long overdue to admit our failure at teleonomic pursuits (itself an oxymoron). The facts of *controls* rather than mere constraints speak for themselves.

Cybernetics deals with steering and control mediated by communication. Biosemiosis provides that communication. But biosemiosis requires linguistic-like rules, not laws. These rules cannot be forced. Their use must be truly contingent, and they must be actively selected. Constraints not only are not purposefully chosen, they have never been observed to produce sophisticated formal function. The only thing ever observed to produce sophisticated functionality is controls, not constraints (Abel, 2010).

All known life is programmed and cybernetically processes computation. The programming of life is a cybernetic function. It prescribes, steers and controls life's formal processes. The entire field of genetics, circRNA regulation, and the use of CRISPR-Cas3, CAS9 and Cas12 technology proves the cybernetic and programming nature of life's controls. Life is clearly not just chemistry. Life is instructed and controlled by genetic and epigenetic Prescriptive Information (Abel, 2009a; D'Onofrio et al., 2012a, b), a special case of Szostak's "functional information" (Hazen et al., 2007; Szostak, 2003), not mere Shannon entropy.

Prescription using representational material symbol systems is formal, not physical—as formal as mathematics (Abel, 2011e, 2024a, b, c). The prescription of formal function is instantiated into physicality using physical symbol vehicles (e.g., actively selected and sequenced nucleosides). These tokens may be physical, but they represent meaning.

Configurable switch-settings are also used to prescribe function (Abel, 2008b, 2010, 2011k, 2012, 2015). Configurable switches must be set in advance of realized integrated circuit efficaciousness. Decision node choices (e.g. nucleoside selection of which of four monomers to

next polymerize to a forming nucleic acid strand) must be active, not passive or secondary (Abel, 2009c, 2011g, k, l, 2015, 2024a, c; Abel and Trevors, 2005; Durston et al., 2007). Such programming and computational processing is formal, not physical.

Physicodynamic assembly would not be formally controlled. It would be merely constrained. Neither chance nor necessity can explain the cybernetics of life.

Heritable linear digital prescriptions use Material Symbol Systems (Abel, 2011g; Rocha, 1995a, b, c). This is the very thing that makes neoDarwinian evolution possible. Prescription of formal function is a different modality altogether from the prescribed physical product itself (Abel, 2011j). The same is true of the physical objects (the tokens or physical symbol vehicles) used to instantiate that prescription of formal function (Abel, 2011g; Esposito et al., 2024).

Nucleotides and triplet codons are formal representational physical symbol vehicles used to prescribe formal meaning and function (Abel and Trevors, 2006a, 2007). So are the multidimensional translational-pausing sextet codes superimposed on the triplet amino acid sequencing coding (D'onofrio and Abel, 2014). So are epigenetic switch-settings (Hernday et al., 2002).

This manuscript may be recorded into cyberspace or recorded with paper and ink molecules. But its content is abstract, conceptual, non-physical and formal. DNA, RNA, chaperones and other useful polypeptides may be physical, but the formal Prescriptive Information instantiated into these molecules is the real issue.

Since all known life is programmed and cybernetically processed by nanomachinery, Turing's halting problem is very relevant when AT purports to measure causation difficulty. If we are to measure the challenge or difficulty of achieving computation, we are going to need to measure the causative techniques of cybernetics, not mere thermodynamic probabilism.

So, the fundamental question is, "Can AT really quantify this *active selection cybernetic process difficulty*?"

2.5. Measurement limitations of AT

AT contends that it can measure the difficulty of biotic molecular formation. If this were correct, it might make possible quantifying the difficulty not only of biomolecular synthesis, but of molecular evolution and possibly even biochemical pathway orchestration independent of any active selection.

AT's assembly index does suggest some after-the-fact difference between biotic and abiotic compounds. But, as pointed out in Section 2.1.1, this difference most likely relates more to molecular length than whether or not any molecule is biotic. Of course, no molecule is living. Many abiotic organic molecules exist, especially in association with hydrothermal vents or where olivine-rich magmatic systems get cooled down and hydrated (Andreani et al., 2023).

"Systems biology" is incredibly orchestrated in its concepts, organization, pathways and methods. Sooner or later, molecular evolution and even neoDarwinian evolution has to be included in the measurement of causation of biologically useful ensembles. Thus, for AT to put a dent into the abiogenesis problem, it must ultimately quantify both physicodynamic and selection challenges in its measure of causation difficulty. And the selection challenge measured must be *active selection* challenge, not passive after-the-fact selection challenge (Abel, 2009c, 2011f, 2024a). Can active selection be measured at all with nothing but entropy measures?

The requirements of biotic molecules to do their specific needed tasks seem inexhaustible. The causation difficulty for bifunctional molecular assemblies is far, far greater than for non-functional stochastic ensembles.

- a. Wouldn't some selections be more challenging to make than others?
- b. Wouldn't some active selections be more consequential than others?

- c. Does AT address any engineering-like *steering or controls* that might be needed to generate function, as opposed to the mere mindless constraints of inertness (Abel, 2010)?
- d. Are any *active selections* at all required to generate assemblies with sophisticated functions? The entire engineering field might tend to question whether natural science could address such active selections in pursuit of utility.

Ultimately, abiogenists wonder how useful AT quantifications will be in measuring the challenge of achieving formal organization, "wise" decision-node selections, integration of circuits, quality control, minimization of required steps to computational success, orchestration of disparate biochemical pathways into homeostatic protometabolism and metabolism, the molecular machinery needed to process the prescriptive information of programming and cybernetic controls, the difficulty inherent in Turing's "Halting Problem" (Church, 1936/Turing, 1936 #7143). The *specific steering and controls* needed to orchestrate protometabolic function cannot be measured with fixed entropy units.

What exactly is AT's "assembly space?" How does it differ from a physical phase space? Is it not the sum of all possible physicodynamic outcomes? Is this not directly analogous to and a measure of Shannon Uncertainty and entropy?

2.5.1. The utter inadequacy of Shannon Entropy measures

Randomness has never programmed any non-trivial computation. Probabilism cannot program. Measurements of phase space cannot program. If programming were done by constraint or law, there would be no contingency. Programming is impossible when constraints and law preclude contingency. Not just contingency, but purposeful choice contingency is the key to cybernetics and the engineering of nontrivial function.

Shannon realized early on, much to his chagrin, that it was impossible to measure purposeful choices. Every choice involved in algorithmic optimization is unique and highly contextual. Each choice has a different weight of importance and criticality to the success of the formal process. Thus, specific positive choices cannot be measured with fixed units. He was forced to settle for measuring Uncertainty, not Prescriptive Information (instructions or controls) (Shannon, 1948a, b; Shannon and Weaver, 1949).

Not even passive, after-the-fact mutation value can be measured with fixed units. Nearly all mutations are either deleterious or neutral. But how could the relative degree of "deleterious" be measured? How could the relative worth of an extremely rare questionably "good" mutation be measured? Some point mutations have major effects. Others have minor or seemingly no effects. That being the case, it is impossible to measure a particular mutation's relative effect with fixed units.

Shannon got irritated when his Entropy measures were called "information" (Shannon and Weaver, 1949). None of the humanly-generated epistemological concepts of differential entropy, conditional entropy, joint entropy, mutual entropy, directed entropy, conditional mutual entropy, or relative entropy have any direct relevance to measuring Szostak's Functional Information (Hazen et al., 2007; McIntosh, 2006; Sharov, 2009, 2010; Szostak, 2003) or Abel's more refined Prescription Information (Abel, 2008b, 2009a, 2010, 2015; D'Onofrio et al., 2012a).

Use of the words "challenge" and "difficulty" by AT authors acknowledges that the formation of these useful molecular assemblies was not straightforward. That challenge is not just probabilistic. Determinative factors are needed. Physicochemical interactions producing formal functions require steering and control toward utility. Thus, the naturalistic goal of AT ought to include the calculation of the formation challenges of uniquely *worthwhile* ensembles forming spontaneously. In abiogenesis research, only "usefulness" would contribute to naturalistic notions of "self-organization" or "emergence" of protocells.

All nucleosides are polymerized with the same 3'5' phosphodiester bond forming difficulty. This has nothing to do with the difficulty of

prescribing polyamino acid sequencing so as to instruct a functional protein or circRNA production. And we are not even addressing the problem of multidimension coding's such as the superimposed translational pausing coding needed for proper protein folding (D'Onofrio and Abel, 2014). A sextet code prescription is superimposed on the triplet coded prescription, providing a completely different class and multi-dimensional set of instructions in a different language. That's formalism, not mere physico-dynamics. It's rule-based, not law-based. Obedience to rules is voluntary. Failure to obey rules typically results in loss of function.

Prescriptive Information was required to instruct the generation of sophisticated functionality in a prebiotic environment. This requirement of orchestration and programming is by far the most plaguing problem of abiogenesis science (Abel, 2015). Of interest to abiogenists is the ontological cause of utility within an inanimate context.

Shannon theory (itself conducted only by human agents) gives us the total number of binary-equivalent options (possibilities) from which an ideal path *could be* chosen. Shannon's measure of Uncertainty does not measure the optimal choices that get us to the destination the fastest, or with the greatest computational efficiency. Shannon theory makes no purposeful choices. Shannon theory tells us nothing about how an algorithm got optimized. The AT measurement is bogus. This supposedly naturalistic model is not naturalistic.

Shannon's "reduced uncertainty" measurements are calculated after-the-fact of having achieved computational halting through wise programming choices by agents. It represents the difference between all possible "steps" and the specific steps that produced the history of successful computation. AT is blind to which happenstantial "choices" from among real computational options will halt. Again, that requires formal cybernetics, not inanimate physical law, constraint measurements, or possibility space.

We can measure Shannon's Uncertainty to measure probability space. But that has nothing to do with function. An agent must use Reduced Uncertainty which first requires known halting Prescriptive Information to subtract from the statistical phase space. Even then, the measurement is only a measure of reduced uncertainty, not a measure of the specific Prescriptive Information choices that instruct halting success most efficiently.

Reduced Uncertainty also relates only to the solution of a *specific problem of interest*. An inanimate environment has no problems of interest. All of this requires agency, not just physycodynamic interactions.

The bottom line is that what is being appealed to is really nothing more than a mere phase/probability space, not a cybernetic (steered toward utility) space. What is actually being sold is a supposedly naturalistic "algorithmic optimization" that in reality does not exist. A mere probability space cannot generate algorithms, achieve sophisticated assemblies, meaningful messages, or computation.

How exactly are the required purposeful choices being produced or measured? Only agents care about achieving a goal with the fewest number of operations. That is called cybernetics, not natural science! Laws and constraints can do no such thing.

The authors are proposing to *measure* the degree of causation difficulty of molecular ensembles. Says Sharma et al.,

"We introduce a *measure* called assembly (A), capturing the **degree of causation** required to produce a given ensemble of objects." (Sharma et al., 2023)

Is it fair to ask for clarification of the nature of these ensembles? Are these molecular ensembles all-inclusive? Are they all of equal value as stochastic ensembles? Do they include highly improbable and highly functional proteins, for example?

Axe found that the overall prevalence of protein sequences performing a specific function by any domain-sized fold may be as low as 1 in 10^{77} , adding to the body of evidence that functional folds require highly extraordinary sequences (Axe, 2000, 2004).

Do these molecular assemblies include exons, highly functional

introns, over 100,000 Circular RNA's (circRNAs) (Liu et al., 2023; Liu and Chen, 2022a, b), microRNA sponges and regulators (Hansen et al., 2013; Memczak et al., 2013), back-splicing (Welden et al., 2022), etc., with all of their instantiated linear digital nonphysical Prescriptive Information (D'Onofrio et al., 2012a, b)? Do these "assemblies" include the specific locations of all the methyl or acetyl groups at certain positions to effect epigenetic controls? Did the "objects" record the pathway for their future synthesis?

AT authors constantly appeal to "discovery."

- What did the "discovering," chance or necessity?
- What did the active-selection programming?
- What equipment processed the programming?
- What took notes and recorded all of the successful decision-node choices that could lead to future needed computational haltings?
- How did inanimacy select for function over non function?
- Did the four known forces of physics care whether the "assembly" achieved highly integrative homeostatic metabolic function rather than useless tar?
- What is the difference in quantification between holistic *formal* success or failure?
- How does the Assembly Index quantify the difficulty level of accomplishing even any of these specific tasks?

For AT to put a dent into the abiogenesis problem, it must quantify both physycodynamic and evolutionary challenge. The "causation difficulty" of biotic molecules would likely be quite different from that of abiotic molecules.

Zenil et al. in three very recent 2024 papers continues to demonstrate that AT lacks empirical evidence that it is different from or outperforms other complexity indexes in connection with selection for molecular utility and evolution (Abrahão FS et al., 2024; Ozelim et al., 2024) Zenil contends that AT proponents are promoting hyperbolic claims.

2.5.2. Can prescriptive information be measured?

Say AT authors, "The 'assembly index' is the number of steps on a shortest path producing the object."

- How was the shortest path determined?
- Was the shortest path physycodynamically determined, or was it chosen?
- Did the "objects" do the choosing?
- Did randomness actively "select" the shortest path to success?
- Did constraints or the laws of physics choose the shortest path to success?

No scientific justification exists for believing that stand-alone ontological physycodynamic factors pursued the goal of "the shortest path." Chance and necessity could not have generated or optimized *process efficiency*.

In reality, the "degree of causation difficulty" measurement would be the degree of cybernetic/computational difficulty. That would be a measurement of engineering difficulty, not a measurement of spontaneous physycodynamic difficulty. But, of course, the Prescriptive Information needed to instruct what path to take is not directly measurable, as Shannon realized. Even after the specific halting computational path is enumerated, that particular path still lacks a direct measurement of effectual choices. Shannon Reduced Uncertainty does not instruct what choices specifically compute.

Turing's halting problem is found to be inherent in this "degree of difficulty" quest. Can the laws of physics and chemistry outsmart Turing's halting problem? Intelligent programmers can't. Measurements of physycodynamic interactions cannot quantify cybernetic/computational/engineering function.

AT authors acknowledge that "Molecules with a high assembly index are very unlikely to form abiotically, and the probability of abiotic

formation goes down as the value of the assembly index increases.” But computational and cybernetic success is far more problematic than that. The Universal Plausibility Metric ξ is way less than 1.0 for spontaneous formation of most any sophisticated protein (Abel, 2009e, d, 2011c, i). No plausible scientific explanation exists within this toy model.

Hazen et al. (Hazen et al., 2024) disagree with the contention that only life can generate molecular structures with an MA index (Molecular Assembly measure) ≥ 15 . Both Hazen et al. and Zenil take issue with AT’s ability to differentiate between living and nonliving systems using single scalar values.

2.6. Human conceptualization limitations of AT

AT “reimagines the concept of matter” within assembly spaces,” (Sharma et al., 2023).

What exactly does our human “imagining” and “conceiving” today have to do with historical ontological Abiogenesis (what actually happened in an inanimate environment back then)?

- a. What did the imagining and conceiving in a prebiotic environment?
- b. Did the physical “objects” do the imagining, or the AT imaginers about the objects?
- c. Does this imagining constitute a physical ontological cause of purely physicydynamic effects?
- d. Is this truly a naturalistic prebiotically plausible model?”

The imaginings of AT theorists are being manufactured not by “possibility space” or “eons of time,” but by purposeful agent choices of possibilities made in order “to make our model work for us.” The “reconceptualizing” of human investigators did the choosing of the shortest, most efficient path, and then measured the shortest steps producing each biotically useful assemblies.

3. Experimental data and controls

Can there possibly be any more ideal data “sample size” than the entire history of human observation? What does this ideal sample size

tell us about the hypothesis of “self-organization” and “emergence” of highly optimized, sophisticated, uniquely functional “molecular assemblies”?

We could calculate a very crude total number of observations in human history. Crude though it may be, that number would nonetheless certainly establish an excellent sample space and source of controls for all known human observations.

The Population Reference Bureau (PRB) suggests that approximately 108 billion people have lived on earth. That’s 1.08×10^{11} people. Suppose each person averaged only a hundred observations per day. That would be 36,500 observations per year per person, or 3.65×10^4 . If the average life expectancy was only 50 years, that would be (3.65×10^4) times (5×10^1) per each person’s lifetime.

$(1.08 \times 10^{11}) \times (3.65 \times 10^4) \times (5 \times 10^1) = 1.971 \times 10^{17}$ = Total Observations. Most of these would not have been recorded. But one recorded observation can summarize the history of many observers and observations. We’ll tentatively use 10^{15} observations as a very crude, relative number of all human observations throughout history.

The hypothesis inviting falsification in this paper is the following: “None of the “molecular assemblies” listed in Table 1 has ever been observed to “self-organize” or “emerge” spontaneously from inanimate nature in the history of human observation.”

4. Results

Suppose no “molecular assembly” theorist can cite a single incident of spontaneous “self-organization” or “emergence” of any one of these entities in this table, independent of active agent selection.

The hypothesis not only stands unfalsified, but should immediately rise well above theory status to virtual paradigm level. The total absence of any such observations in human history affirms in a powerful way the fundamental F > P Principle (The Formalism > Physicality Principle) (Abel, 2011e). This Principle emphasizes Formalism’s dominion and control over physicality. The F > P Principle defines the ontological primacy of Formalism and its governance of Physicality. Physicality is governed by Mathematical laws. We dichotomize natural science from Engineering for good reasons.

Table 1
Some simple, common “molecular assemblies” never having been observed in human history to come into existence apart from agent active selection, steering and control with the aim of achieving usefulness.

Simple Molecular Assemblies with uniquely valuable functions:	Controls: Huge sample size of all human observations in recorded history	Total number of human observations of self-assembly/self-organization/emergence of each of these entities in recorded history	% of spontaneous emergence out of total # of historical observations	Universal Plausibility Metric for the hypothesis of this “emergence” and “self-organization”
Paper clip	$>10^{15}$	0	0	$\xi = 0$, or <1.0
Pencil	$>10^{15}$	0	0	$\xi = 0$, or <1.0
8 ½ X 11clean white sheet of paper	$>10^{15}$	0	0	$\xi = 0$, or <1.0
A Highlighter	$>10^{15}$	0	0	$\xi = 0$, or <1.0
Wooden clothespin	$>10^{15}$	0	0	$\xi = 0$, or <1.0
Rubber stamp saying “Filed”	$>10^{15}$	0	0	$\xi = 0$, or <1.0
Stapler	$>10^{15}$	0	0	$\xi = 0$, or <1.0
Music stand	$>10^{15}$	0	0	$\xi = 0$, or <1.0
Clock	$>10^{15}$	0	0	$\xi = 0$, or <1.0
Printer	$>10^{15}$	0	0	$\xi = 0$, or <1.0
Suspension Bridge	$>10^{15}$	0	0	$\xi = 0$, or <1.0
Guggenheim museum building	$>10^{15}$	0	0	$\xi = 0$, or <1.0
Mars rocket	$>10^{15}$	0	0	$\xi = 0$, or <1.0
Multi-step biochemical pathway	$>10^{15}$	0	0	$\xi = 0$, or <1.0
Representational sign/symbol system	$>10^{15}$	0	0	$\xi = 0$, or <1.0
Molecular machines	$>10^{15}$	0	0	$\xi = 0$, or <1.0
Subcellular Nanocomputers	$>10^{15}$	0	0	$\xi = 0$, or <1.0
Subcellular Computation	$>10^{15}$	0	0	$\xi = 0$, or <1.0
Homeostatic metabolism far from equilibrium in a hostile environment	$>10^{15}$	0	0	$\xi = 0$, or <1.0

A simple paper clip comes into existence only by active selection, not by raw physiodynamics. Active selection is a formalism. Previous research has demonstrated that active selection is absolutely essential even for molecular evolution to occur (Abel, 2024a).

The Data is perfectly clear. The results indicate the inadequacy of what natural science has always labeled “natural causation” to generate any of these formally motivated and enabled molecular assemblies. These are all molecular assemblies specifically tailored to perform needed functions to which the laws, constraints and forces of physics and chemistry are blind. In short, these entities, devices, tools, machines as effects have never been observed to come into existence from “Physicodynamic Causation” alone. Some additional factor is required for their generation into existence.

5. Discussion of results

The Universal Plausibility Principle requires peer review rejection of any hypothesis in any field of natural science that measures out with a Universal Plausibility Metric of $\xi = < 1.0$ (Abel, 2009e, 2011i). The hypothesis that any one of the effects in this table was caused by Physicodynamics alone is fully negated by the Universal Plausibility Principle published in 2009 (Abel, 2009d, 2011i) No letters to the Editor, no conference debate objections, and no sequentially published criticisms of this natural science-wide general Principle of Scientific Method Analysis has ever arisen in 15 years.

Even if the estimate of total number of human observations in history were greatly lowered to only 10^{10} , the results would be the same. The Universal Plausibility Metric ξ would still be below < 1.0 , requiring peer-review rejection of the self-organization/spontaneous emergence hypothesis as a respectable model for plausible consideration as a respectable scientific hypothesis.

Belief that any one of these molecular assemblies in the table self-organized and emerged spontaneously (naturalistically from Physicodynamics alone) is scientifically implausible, and should be rejected by any Editor and reputable peer review. Assembly theory is bogus even before trying to measure the level of difficulty.

What accounts for none of these molecular assemblies ever having been observed to “organize itself into existence”? Why has not a single one of these “molecular assemblies” ever spontaneously “emerged”? The answer is that all of them require steering, controls (not mere constraints (Abel, 2010), a valuing of functionality, aims, goals, and engineering-like decisions to bring them into existence). Physicodynamic laws and constraints alone cannot generate them. The four known forces of physics and thermodynamic possibilities are inadequate to explain their causation, existence and utility.

6. Conclusions from the data

How did any molecular assembly in Table 1 come into existence, then? Only by outright engineering choices. No other form of causation other than *Choice Causation* can cause the effect of an algorithmically optimized paper clip.

Highly tailored functionality cannot be optimized in molecular assemblies without active selection at the time of assembly. Algorithmic optimization is impossible without Choice Causation. Active Selections in pursuit of function had to be made prior to the existence of any of those functions. Function not only had to be anticipated, but pursued through decision-node choices and particular configurable-switch settings needed to integrate circuits.

Paper clips, pencils, 8 ½ X 11 clean white smooth pieces of paper, Highlighters, clothespins, and everything else in Table 1 came into existence only via controls, not constraints or laws. Algorithmic optimization decisions and choices originate only from the far side of the Cybernetic Cut, not the near side (Abel, 2008b, 2011b). Very real Choice Causation was conducted into physicality via the one-way Configurable Switch Bridge that alone traverses the infinitely deep ravine between

formalism and physicality. The mathematical laws of physics, for example, emanate from the far side of the Cybernetic Cut. Nothing on the near physical side can influence anything on the far side of the Cybernetic Cut. The configurable Switch Bridge is a narrow one-way bridge over the infinitely deep ravine that divides formalism from physicality (Abel, 2008b, 2011b).

Self-organization is a nonsense term and concept that is self-contradictory. “Self-organization” and “emergence” are notions that have absolutely no place in natural science, and constitute a non-viable hypothesis that should be rejected by editors and peer review in any natural science journal.

7. Overall discussion

Jonathan Bartlett makes a valiant effort to sort out teleonomy from teleology in “Causal Capabilities of Teleology and Teleonomy in Life and Evolution” (Bartlett, 2023b) The whole point of teleonomy is to naturalize teleology. Teleology can only originate from the far side of the Cybernetic Cut (Abel, 2008b, 2011b, 2024a, b, c). Teleonomy does not explain how physiodynamics alone on the near side could have pursued or generated the formal programming choices needed for cybernetic computational success.

Life is fundamentally formal. Life is ongoing computational success, otherwise known as correlated, integrated cybernetic haltings. The latter is every bit as formal as mathematics, logic theory, language, and representational symbol systems. None of these aspects of reality can be reduced to mere physico-chemical phenomena.

Time, including four billion years of time, is not a cause of any effect! Lee Cronin says the mechanism was four billion years of “technology”—“four billion years of search, destroy and survive.”—(Cronin, 2023) Are “search” and “technology” natural science terms? Do they have naturalistic meaning, or formal engineering/cybernetic meaning? How exactly did an inanimate, prebiotic environment “search” for or achieve “technology”? Can we not see how utterly fallacious is the entire pre-supposition of a purely physicalistic orchestration, assembly and spontaneous generation of life? Even “trial and error” constitutes a *search mechanism*. An inanimate environment conducted no searches!

The most fundamental need of life-origin research is to elucidate the source of orchestrational prowess needed to prescribe and formally organize even the simplest imaginable homeostatic proto-metabolism (Abel, 2011b, d, e, f, h, k, 2012, 2015). It requires explaining the mechanisms of generating the eventual cybernetic processing of that programming by sophisticated machinery. Biochemical pathways must not only be prescribed; they must be formally processed by subcellular nanocomputers. Molecular machine utility must be orchestrated into highly conceptual integrated circuits. Add to this the myriads of synthetic biochemical problems (Thaxton et al., 2020; Tour, 2020, 2024).

Does AT really measure the steering and control needed to craft the primary, secondary and tertiary structure of highly efficacious enzymes? If AT pre-assumes nothing more than chance and necessity, ignoring the third fundamental category of reality (Active Selection) (Abel, 2011b, 2024a), it is guaranteed that it’s measurements of difficulty will be bogus.

The reason “laws do not predict emergence,” as AT authors admit, is that laws cannot produce Choice Causation in pursuit of formal function (Active Selection from among real physiodynamic options).

Successful measurement of the challenge of functional assembly of each and every physico-chemical sophisticated molecule, and the difficulty of orchestrational integration, should not be naively proclaimed.

AT is little more than the same embarrassing boondoggle of Richard Dawkin’s 1986 *Blind Watchmaker* model of supposed evolution of a linguistically functional phrase (Dawkins, 1986). The model was altogether dependent upon the agent experimenter’s successive active selections. Dawkins chose which phrase to keep building on in the step-wise construction of message meaning. Linguistic rules must be followed in any language. Dawkins was blind to his own investigator

involvement in his experimental design. So are Sharma et al.

Agent Choice Causation and agent Concept are being snuck in through the back door of AT's supposedly naturalistic model. Imagination is being conveniently confused with "history" and "possibilities" in "conceiving" computational success without any basis for active selection. AT ignores Turing's cybernetic halting problem.

Quantum mechanics doesn't even come close to explaining the engineering realities of subcellular life. As is the case with irreversible nonequilibrium thermodynamics, mere probabilism obtains, not any active selection that might cause the effect of formal "usefulness." Quantum mechanics is inseparable from extensive epistemological confusion and misrepresentations of pseudo-engineering prowess.

AT cannot measure the orchestrational challenge of even a theoretical protometabolism. Life is not just a "molecular assembly." Physico-chemically, assembly is nowhere near as likely as disassembly. Life is a highly integrated formal *system* of innumerable cybernetic computational operations. Life performs innumerable sophisticated holistic functions in concert through time far from equilibrium. Life consists of Sustained Functional Systems (SFS) (Abel, 2012b), not just thermodynamic momentary states. Though instantiated into physicality, life is fundamentally formal rather than physicydynamic.

Life is the ultimate in algorithmic optimization. Algorithms cannot be optimized without active selection. Selections must be made "in order to..." Such choices cannot be attributed to evolution. Choices at true decision nodes must be made prior to the realization of any secondarily favored phenotypic function of already-living organisms.

8. Conclusions

"AT" does not appear to be a naturalistic prebiotic cause-and-effect model that will benefit abiogenesis research. Present-day reconceptualization's of physicality in the minds of scientist agents have little relevance to what actually happened physicydynamically in a prebiotic environment. Abiogenists need stand-alone models of self-organization and emergence that can be empirically verified without investigator involvement in experimental design.

AT authors have erroneously promoted AT's capability of differentiating biological from inanimate physical phenomena. No real divergence of Assembly Theory from LZ compression exists in the model. AT authors' claims are very overstated based on the data and methods of analysis. Instead, AT shows asymptotic convergence to mere LZ compression and Shannon Entropy (Abrahão FS et al., 2024; Ozelim et al., 2024; Uthamacumaran et al., 2024; Zenil, 2023, 2024). Nothing new exists in AT quantification approaches relevant to distinguishing the formation of life from nonlife. Their AT assembly index reduces to nothing more than a special case of Zenil's decade-old block Decomposition Method. AT simply reshapes elementary principles of complexity that have nothing to do with abiogenesis.

In reality, empirical evidence of formal "self-organization" or "emergence" is sorely lacking. Something would have to already exist in order to organize itself into existence. "Self-organization" is a nonsense term that has no place in scientific literature. Under no circumstances do formalisms ever spontaneously "emerge" from physicydynamics. All formalisms invariably require active selection from among real physicydynamic options if those formalisms are to be instantiated into physicality. Life is first and foremost a constellation of instantiated formalisms. Life is the ultimate in a concert of integrated halting computations (Abel, 2011).

Mere entropic and statistically-based complexity is not synonymous with life. Many inanimate phenomena are complex. Life is *conceptually* complex, not just complex. Conceptual complexity originates only from the far side of the Cybernetic Cut—the great ravine that divides mass/energy physicydynamics from formal active selection, Choice Causation and Prescriptive Information generation (Abel, 2008b). Controls from the far side of the Cybernetic Cut constitute the very hallmark of life. This is clearly established by life's programming with Prescriptive

Information that employs Material Symbol Systems. Representational instructions are instantiated into physicality using physical symbol vehicles. The cybernetic processing of life's programming requires very sophisticated nanocomputers and molecular machines. Not a single one of these highly unique devices has ever been observed to be generated by spontaneous molecular assembly. The same Cybernetic Cut divides engineering from natural science.

AT "measurers" are intelligent choosing agents who conflate mere complexity and "novelty" with homeostatic metabolism in their minds. Formal orchestration is required to organize even a theoretical protometabolism. Neither the physical "objects" (the spontaneous molecular "assemblies" [stochastic ensembles]) nor the fixed laws and constraints of physics and chemistry can account for the hundreds of highly tailored essential players needed for abiogenesis. No basis for orchestration of any bona fide formal organization is presented by AT. Spontaneous homeostatic protometabolism is just religiously and blindly believed to have occurred in accord with the purely metaphysical Kuhnian paradigm rut that pre-assumes axiomatically that "Physicality is sufficient." The cardinal faith and fanatical pontification of philosophic naturalism is that inanimate "Physicydynamics orchestrated life."

Irreversible nonequilibrium thermodynamics has not lived up to its hype. Neither have Complexity, Chaos and Assembly Theories. Molecular evolution is impossible without *active* selection (Abel, 2009c, 2011f, 2024c). If AT, Complexity Theory, Chaos Theory, Irreversible Non-equilibrium Thermodynamics, Quantum Mechanics and Molecular Evolution Theory are collectively so capable of explaining abiogenesis, why have we not seen scores of instances of spontaneous generation of life over the last few centuries? We have never witnessed the spontaneous formation of a simple piece of metal alloy of constant diameter called "wire." What empirical or rational justification exists for believing that homeostatic metabolism just "emerged" from nothing but chance and necessity?

Science, biology especially, must stop excluding the third repeatedly observable fundamental category of reality—Active Selection (Abel, 2011b, 2024b, c)—from its study of objective reality. It must at least be willing to acknowledge the reality of subcellular biological engineering and computation when it sees it. Science requires self-honesty, not obfuscation. The programming, cybernetic processing, computational halting, nanocomputers, and molecular machines employed by even the simplest organisms are all formalisms. Life is controlled, not constrained. Life is rule-based, not law-based. Life is as fundamentally formal, abstract, conceptual and nonphysical as the *mathematical* laws of physics themselves.

Funding

The Origin of Life Science Foundation, Inc. Abe3-2024.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.biosystems.2024.105378>.

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